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## The Motivational Function of Regulatory Focus in Creativity

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**ABSTRACT** The Regulatory Focus Theory maintains that people may focus on achieving positive outcomes (have a promotion focus) or avoiding negative ones (have a prevention focus) when they pursue their goals. Under a promotion focus, people would formulate as many strategies as possible to attain their goal, and hence be fluent in idea generation when they perform a creative task. In contrast, people under a prevention focus would seek to avoid the negative consequences of failing to attain a valued goal, and persist even when the likelihood of success in a creativity situation is small. We tested these predictions in a study, where regulatory focus was measured as an individual differences variable (Part 1) and induced by a goal framing manipulation (Part 2). The results supported our predictions, and suggested that creative accomplishment requires flexible alternation of regulatory foci at the different stages of creative undertakings.

**INTRODUCTION** Creativity is an important building block of human intelligence. Early research on the psychology of creativity has focused on identifying the personality traits of creative individuals, or the cognitive skills involved in the creative process (for review, see Sternberg & Lubart, 1995). Relatively little attention has been given to the role of self-regulation in creative thinking. In the present research, we investigated how self-regulatory focus may contribute to creativity.

Our focus is on ordinary creativity rather than extraordinary creativity or creativity possessed by a few exceptional individuals. We view creativity as a goal-directed process, and assume every person can become more creative by regulating his or

her goal-directed behavior. As a goal-directed behavior, creativity consists of stages such as task identification, idea generation, solution finding, transformation of ideas into products, and outcome evaluation (Amabile, 1996; Isaksen & Treffinger, 1985). It is beyond the scope of the present research to examine the effects of regulatory focus on every stage of creativity. Therefore, we have limited ourselves to the role of regulatory focus in idea generation and in perseverance of creative pursuits in the face of unforeseen obstacles. In the literature, regulatory focus has been treated both as a chronic individual differences variable (a general dispositional propensity) and a momentary motivational orientation induced by the situation (a state). Although people have a general propensity to adopt a particular self-regulatory style, situational factors could move them to adopt alternative styles. To illustrate the role of regulatory focus in creative thinking, we examined (a) how chronic self-regulatory focus may be related to fluency in idea generation, and (b) how situational inducement of regulatory focus may influence perseverance of creative pursuits.

#### SELF-REGULATION AND CREATIVITY

Recently, some authors have written on the important role of motivational factors in creativity. Amabile (1996), a leading creativity researcher, has pointed out the need to consider motivational factors in creative processes. Sternberg and Lubart (1995) also emphasize the importance of motivation in their "confluence approach" to creativity. These authors posit that having the potential for creativity is not enough. To turn creative ideas into creative products, individuals need to be motivated to actualize their creative potentials in response to the changing needs of the environment. In this theoretical context, the regulatory focus theory (Higgins, 1997) may provide a new perspective on the relationship between motivation and creativity.

According to the Regulatory Focus Theory (Higgins, 1997), people differ in their regulatory focus as a result of different experiences of child-caretaker interaction. When caretakers use the presence or absence of positive outcomes to train their child, they are communicating the importance of "attaining accomplishments or fulfilling hopes and aspirations." In contrast, caretakers who use the presence or absence of negative outcomes to train their child are communicating the importance of "insuring safety, being responsible, and meeting obligations" (Higgins, 1997, p. 1282). The former interaction would result in a focus on promotion in self-regulation, which

represents personal goals as ideals, whereas the latter would result in a focus on prevention, which represents personal goals as duties. The promotion focus is associated with an approach strategy to maximize the chance of actualizing one's ideals. In contrast, the prevention focus is associated with an avoidance strategy to minimize the chance of not fulfilling one's duties (Crowe & Higgins, 1997; Higgins, Roney, Crowe, & Hymes, 1994).

Although there is no direct evidence that links regulatory focus to creativity, research evidence shows that promotion focus persons frequently experience cheerfulness-related emotions and eagerness-related motivations. They are relatively likely to take risks and choose many means to attain personal goals. In contrast, prevention focus persons frequently experience quiescence-related emotions and vigilance-related motivations. They are relatively inclined to avoid risks and restrict themselves to a few means to attain personal goals (Crowe & Higgins, 1997; Higgins et al., 2000). These findings suggest that promotion focus individuals are inclined to generate a variety of distinct alternatives, choose many instrumental means to attain personal goals, and have a risky response bias. Accordingly, compared to those who do not focus on promotion, promotion focus individuals may be more fluent in idea generation when they approach a creativity task, although the ideas they generate may not be more original. Consistent with this hypothesis, Liberman, Molden, Idson and Higgins (2001) reported that when explaining social events, people with a promotion focus tend to generate many and simultaneously endorse multiple hypotheses. By comparison, people with a prevention focus tend to generate and endorse only a few hypotheses.

In addition, Shah and Higgins (1997) have successfully induced a promotion focus by framing a goal as an ideal, and a prevention focus by framing the same goal as a duty or "ought." They found that when a promotion focus was primed, participants were highly committed to their goal when they valued the goal *and* at the same time expected a high likelihood of goal attainment. In contrast, when a prevention focus was primed, because participants saw goal attainment as an obligation, they attached high value to the goal. As long as they valued the goal, they were committed to the goal, even when the likelihood of attaining it was not high.

Successful creativity also requires perseverance in the face of obstacles (Amabile, 1990; Sternberg & Lubart, 1995). The findings reviewed above suggest that when the likelihood of

goal attainment in a creative project is lowered due to some unforeseen obstacles, prevention focus individuals are more likely to persevere than promotion focus individuals, as long as prevention focus individuals attach a high value to the goal. Thus, in the face of unforeseen obstacles, a prevention focus should be more effective than a promotion focus in sustaining people's effort in the creative project.

The foregoing analysis implies that successful creativity requires individuals to shift their regulatory focus in the creative process. A promotion focus may be more useful than a prevention focus at the stage of idea generation. We predicted that a promotion focus would be positively associated with fluency in idea generation, but not necessarily with the originality of the ideas generated. By contrast, a prevention focus would not be associated with fluency and originality. However, when unforeseen obstacles lower the expectancy of goal attainment, a prevention focus would lead to greater commitment to the creative project than would a promotion focus.

As noted, individuals possess both self-regulatory foci, although there are chronic individual differences in the relative propensity to adopt one or the other regulatory focus. Accordingly, situational factors can prime or temporarily induce one focus or the other (Higgins, 1998). As Higgins (1999a) puts it, "an individual with low chronic accessibility in a priming situation could function like an individual with high chronic accessibility in a no-priming situation" (p. 85). In Part 1 of the present research, we examined the role of regulatory focus in the generation of creative ideas by measuring chronic individual differences in regulatory focus. In Part 2, we experimentally manipulated regulatory focus and observed the manipulation's effects on perseverance in the face of obstacles.

### Part 1: Chronic Self-Regulatory Focus and Generation of Creative Ideas

**METHOD** *Participants.* The participants were 35 undergraduate students (29 women and 6 men) in a university in Hong Kong, with a mean age of 21.1.

*The Regulatory Focus Questionnaire.* We measured promotion and prevention foci by the Regulatory Focus Questionnaire (RFQ; Higgins et al., 2000). Higgins et al. (2000) found that the questionnaire has high internal reliability ( $\alpha = .73$  to  $.80$ ). They have also reported extensive evidence pertinent to

the validity of the measure. Examples of the promotion focus items are: “When it comes to achieving things that are important to me, I find that I don’t perform as well as I ideally would like to,” and “Do you often do well at different things that you try?” Examples of the prevention focus items are: “How often did you obey rules and regulations that were established by your parents?” and “Growing up, would you ever cross the line by doing things that your parents would not tolerate?”

Ip and Chiu (2000) selected five items with factor loading greater than .50 on the promotion scale and 4 items with factor loadings greater than .50 on the prevention scale in the Higgins et al. (2000) study, and translated them into Chinese. In a validation study (Ip & Chiu, 2000), we have successfully replicated the factor structure of the nine items reported in the Higgins et al. (2000) study. The two factors (promotion and prevention) accounted for 32% and 18% of the total variance, respectively, in the Chinese version and had good internal reliability ( $\alpha = .68$  for the promotion scale, and  $.79$  for the prevention scale). The two scales also correlated with other measures of personality and cognitive styles in a theoretically meaningful manner in this validation study, as they did in the Higgins et al. (2000) studies.

The participants responded to the 9 items on a 5-point Likert scale, from 1 (“never or seldom”/“certainly false”) to 5 (“very often”/“certainly true”). Some of the items are phrased in the form of a statement. For these items, participants rated how true each statement was to themselves. In calculating the promotion focus and the prevention focus score, some items were reverse scored.

*Idea Generation Task.* After the participants had completed the RFQ, they read a short passage that described pink dolphins and their importance to Hong Kong. They were told that the pink dolphin was a lively and lovely animal. However, due to the Government’s mindless exploitation of the marine environment, they faced the threat of extinction.

Next, participants were asked to generate creative ideas to raise funds and to increase the public’s awareness of the hardship faced by the pink dolphins. They were given 15 minutes to produce as many ideas as possible, and then rated each idea they had generated on its level of creativity on a 7-point scale (1 = “absolutely not creative” and 7 = “very creative”). They also selected two ideas they considered to be most creative. A major dependent variable was the fluency score, which was the number of ideas generated by each participant.

We used Amabile's (1996) Consensual Assessment Technique (CAT) to evaluate the originality of the participants' ideas. In CAT, the originality of creative ideas is determined by a panel's global ratings of the ideas based on the operational definition of originality supplied by the researcher. The recruitment of judges is not limited to experts in a specific domain. In the present study, two psychology graduate students, two experts from the marketing profession, and two non-experts from other professions rated the originality of the ideas produced by the participants. They rated each idea on a 7-point scale (1 = "absolutely unoriginal" to 7 = "very original"). The judges were blind to the participants' scores on the RFQ.

Each participant received an originality score on each idea generated. For each participant, we added the originality scores for all the ideas received from the same judge and divided the sum by the number of ideas generated. This resulted in six aggregate originality scores for each participant (one from each judge). The effective inter-judge reliability ( $r = .96$ ) and the judge-to-judge reliability ( $r = .78$ ) were high. Thus, for each participant, we constructed an overall originality score by averaging the six scores.

## RESULTS AND DISCUSSION

Table 1 shows the zero-order and partial correlations between the two regulatory foci and the two creativity scores. The correlation between the promotion scale and the prevention scale was not significant,  $r(33) = .04, p = .81$ . As expected, the promotion scale had a significant positive correlation with the fluency score,  $r(33) = .45, p < .01$ , and the prevention scale was not related to fluency in idea generation. There was a reliable difference between the size of the partial correlation between the fluency score and promotion focus and that between the fluency score and prevention focus,  $t(32) = 2.12, p < .05$ . The originality index was not related to either regulatory focus. To provide a more stringent test of the association between promotion focus and idea fluency, we performed a multiple regression analysis on the fluency score, with the originality score, the prevention score, and the promotion score as predictors. The association between promotion focus and the fluency score remained reliable ( $b = 2.25, t(31) = 2.73, p = .01$ ), after controlling for the effects of originality and prevention focus. The originality ( $b = .13, t(31) = .15$ ) and prevention scores ( $b = -.04, t(31) = -.08$ ) did not predict idea fluency.

We then used a median split to dichotomize the promotion and prevention scores, and performed a 2 (Promotion) X 2

(Prevention) analysis of variance on the creativity scores. For the fluency score, there was a significant promotion main effect,  $F(1,31) = 5.81, p < .05$ . Individuals with relatively higher promotion scores generated more ideas ( $M = 7.86$ ) than did those with relatively lower promotion scores ( $M = 6.07$ ). The prevention main effect and the interaction were not significant,  $F_s < 1$ . We did not find any significant effects on the originality score, and mean originality scores in the four Promotion X Prevention groups were very similar (range = 3.54 to 3.89).

TABLE 1. Zero-Order and Partial Correlations Between the Promotion and Prevention Scales of the RFQ and the Creativity Indices.

|             | Zero-order correlation |            | Partial correlation |            |
|-------------|------------------------|------------|---------------------|------------|
|             | Promotion              | Prevention | Promotion           | Prevention |
| Fluency     | .45*                   | .01        | .45*                | -.01       |
| Originality | .16                    | .01        | .16                 | .01        |

Notes.  $N = 35$ . Correlation between the Promotion and Prevention Scales = .04, ns; \*  $p < .01$

Recall that the participants also rated the creativity of each of the idea they generated. For each participant, we summed the creativity self-ratings of the generated ideas and divided the sum by the number of ideas generated. Participants with a strong promotion focus rated their ideas as more creative than did those with a relatively weak promotion focus,  $r(35) = .41, p < .05$ . Prevention focus was not related to the participants' own ratings of their creativity,  $r(35) = -.02$ , ns. In addition, the participants' own ratings of their creativity were not related to the fluency score ( $r = .03$ ), or to the judges' ratings of originality ( $r = .13$ ).

We also examined the correlations between self-ratings of creativity and the creativity scores among participants with a predominant promotion focus and among those with a predominant prevention focus. In this analysis, we operationalized predominance in regulatory focus by a median spilt of the difference between the promotion and prevention scores. Based on this operation, predominant promotion focus participants had a relatively higher promotion score than prevention score, and predominant prevention focus participants had a relatively higher prevention score than promotion score. Among both groups of participants, self-ratings of creativity did not correlate with idea fluency ( $r = .21$  for the predominant promotion

participants, and  $r = -.22$  for the predominant prevention participants), or with idea originality ( $r = .17$  for predominant promotion participants, and  $r = .09$  for the predominant prevention participants). Thus, despite the fact that the participants' self-ratings of creativity were not related to their creativity scores, promotion focus individuals displayed a positive bias in the evaluation of their own creative ideas. It appears that for promotion focus individuals to maintain a high expectancy of success in achieving positive outcomes, they need to believe that they have formulated many good ideas.

In short, consistent with previous findings (Higgins, 2000), the correlation between the two regulatory focus scales was low, indicating that the two regulatory foci are independent constructs. As in Amabile (1996), there was good agreement between judges on the relative originality of the ideas generated by the participants, as indicated by the high inter-judge reliability.

The results also supported the prediction that promotion-focused individuals would generate more ideas than less promotion-focused individuals in a creative task. In contrast, prevention focus was not related to fluency in idea generation. However, despite the positive correlation between promotion focus and the fluency score, promotion focus was not related to the originality score. Thus, promotion-focused individuals seem to have a stronger motivation to generate more means to solve a creative problem, but their ideas may not necessarily be original.

## Part 2: Induced Regulatory Focus and Persistence

**METHOD** Immediately after the participants had completed Part 1 of the study, we randomly assigned them into the promotion focus or the prevention focus group. The participants in both groups were asked to take the perspective of the protagonist in a hypothetical story. In the story, the protagonist was pursuing a valued goal, but the goal was framed differently in the two experimental conditions. In the version of the story read by the promotion focus group, the protagonist was trying to achieve positive outcomes, whereas in the version read by the prevention focus group, the protagonist was avoiding negative outcomes. The story presented to the two conditions were:

*Promotion focus.* You are a Grade 10 student who wants to join the advertising industry. This is your ideal career, and you want to maximize the chance of getting

*into the profession.* Therefore, you plan to enter a competitive professional school. This school admits students based on both academic performance and participation in creative activities. Past admissions show that *a good result in an annual advertising campaign competition would increase your chance of being accepted by the school.* Therefore, you enter the competition. You know that *having good performance in the competition will increase your chance of being admitted to the professional school.*

*Prevention focus.* You are a Grade 10 student who wants to join the advertising industry. *You worry that you cannot realize your goal, and want to minimize the chance of not being able to enter the profession.* Therefore, you plan to enter a competitive professional school. This school admits students based on both academic performance and participation in creative activities, including participation in an annual advertising campaign competition. Past admissions show that *those who do not join the competition or those who do not do well in it are likely to be rejected by the school.* Therefore, you enter the competition. You know that *having good performance in the competition will lower your chance of being rejected by the professional school.*

To introduce an unforeseen obstacle that would lower the expectancy of goal attainment, all participants read the following ending of the story:

This year, the topic for the advertising campaign competition is “Save the Pink Dolphin.” Now you have the opportunity to use the ideas you just generated to join the competition. However, for some reasons, you cannot use either of your best two ideas to join the competition. Not being able to use your best ideas, your chance of having good performance will decrease.

Having read the story, the participants indicated the probability that they would join the advertising campaign competition if they were the protagonist, on a scale that ranged from 0% (“absolutely not joining the competition”) to 100% (“will certainly join the competition”).

At the end of the experiment, as a manipulation check, we had the participants respond to two items: (a) “How worried are you about not being admitted into the advertising school?”

and (b) “How much do you wish to be admitted into the advertising school?” The participants indicated their response to these items on a 9-point scale from 1 (not at all) to 9 (very much). All participants in the prevention focus group had a score of 5 or above on the first item and a score of 4 or below on the second item, and all participants in the promotion focus group had a score of 5 or above on the second item and a score of 4 or below on the first item. In short, the manipulation was successful.

RESULTS AND DISCUSSION

As expected, there was a significant difference between the promotion focus group and the prevention focus group in the likelihood of joining the advertising competition,  $t(26) = -2.47$ ,  $p < .05$ . Table 2 reveals that the prevention focus group was more likely to join the competition than was the promotion focus group (see Table 2).

TABLE 2. Mean Value and Standard Deviation of Persistence for Participants Framed in Different Regulatory Foci.

|                  | M     | SD    |
|------------------|-------|-------|
| Promotion Focus  | 86.15 | 13.25 |
| Prevention Focus | 96.00 | 7.37  |

As in the Shah and Higgins (1997) study, the present study showed that both the value and expectancy of goal attainment are necessary for a high degree of goal commitment under a promotion focus. In the promotion focus group, when the expectancy of goal attainment decreased, despite that a decision to join the competition had already been made, goal commitment dropped to 86.2%. In the prevention focus group, goal commitment remained very high (96.0%), even when the expectancy of goal attainment was reduced. This finding is consistent with the self-regulatory focus theory, which predicts that under a prevention focus, people are committed to pursue their valued goal even when the expectancy of goal attainment is relatively low. When the goal is framed as an obligation, the motivation to avoid the real or imagined consequences of not being able to fulfill an obligation may drive people to stay on the project as long as there is still a reasonable chance of achieving the goal.

Although the participants in the present study were Hong Kong Chinese, our manipulation checks revealed that the

promotion and prevention focus manipulations were effective. Moreover, once the prevention focus was induced, participants showed the theoretically meaningful response pattern (persistence in goal pursuit). Previous research (Ip & Chiu, 2000) has shown that compared to European Americans, Hong Kong Chinese have a lesser tendency to focus on promotion. However, the motivational processes that characterize the promotion and prevention foci appear to be similar across cultures. Findings from the present study are consistent with this general conclusion.

#### GENERAL DISCUSSION

The findings from the present study suggest that self-regulatory competence may contribute to some component processes involved in creative accomplishment. Such competence requires individuals to flexibly shift between a promotion focus and a prevention focus at the different stages of a creative undertaking. A promotion focus may contribute to ideational fluency. To enhance children's ideational fluency, parents and teachers need to provide children with an environment that nurtures the drive for accomplishments or a focus on fulfilling hopes and aspirations. An environment that emphasizes safety, responsibility, and meeting obligations may not facilitate the development of ideational fluency (see also Csikszentmihalyi, 1996; Koestner, Ryan, Bernieri, & Holt, 1984).

On the other hand, a prevention focus is useful for maintaining persistence in the creative project despite low probabilities of attaining one's goal. Creativity researchers have reiterated the importance of perseverance in the face of obstacle in creative endeavors (Amabile, 1989; Steinberg & Lubart, 1995). Thus, parents and educators should help children to manage their regulatory foci, so that they know when to put aside their promotion focus and to avoid the negative outcomes of failing to attain a valued goal.

As Higgins (1999b) maintains, promotion and prevention form a motivational dual, which is possessed by all people. Socialization can make one of the two outcome focus systems more chronically accessible; however, situational factors can also make one focus or the other more accessible temporarily. Because successful creative pursuits require flexible and discriminative adoption of the promotion and prevention foci, the finding that self-regulatory focus can be situationally induced creates new possibilities for educators to explore the optimal context for enhancing creativity at the different stages of a creative undertaking.

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AUTHOR NOTES

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